Why These Simple Laws? A Mathematical Account of Physics

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Abstract

One rule can explain many things without specifying each separately. This paper develops that idea into Matheism's broad account of physics: simple laws, quantum observations, geometry, time and the development of minds. Under a fixed binary description measure, every extra bit halves a description's initial weight. Laws, starting conditions and observation rules must therefore be counted together; explaining one through another earns a real saving. Quantum evolution without collapse offers one candidate economy. A common generator of particles and measurement settings offers a superdeterminist route to investigate. For time, the paper connects initiated information production with the necessary development of observer organization from a beginning that lacks it. Biological development consumes available energy while total entropy increases. The synthesis also addresses fine-tuning and emergent geometry. Its goal is a complete account in which shared structure replaces independent assumptions; the proposed physical preferences require comparisons of complete models and their observations.

Keywords: Simplicity Dominance Principle, mathematical universe, algorithmic probability, Kolmogorov complexity, many-worlds, superdeterminism, arrow of time, fine-tuning, string theory, consciousness, mathematical monism

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1. Introduction

A few rules can produce a world far more elaborate than the rules themselves. The central physical question is whether this economy explains the universe we observe. Matheism supplies the proposed foundation: matter, laws and observers belong to one mathematical reality. This paper asks what follows when we compare complete accounts of that reality, including their starting conditions and the records observers obtain.

Universal realization is a further ontological commitment. If every consistent structure is physically realized, consistent selection rules are included. The reverse inference does not follow: the consistency of a filter, including the filter that excludes nothing, does not establish its physical realization without assuming the ensemble being argued for. Mathematical monism motivates the proposal but does not by itself prove that every mathematical possibility is physical.

The multiverse proposal raises a measure question: which structures and observer locations should receive what weight? This paper proposes a simplicity prior proportional to $2^{(-K)}$, together with a separate account of observations. Its conditional weight ratios are mathematical facts. The choice and physical interpretation of that prior need their own justification; identifying reality with structure does not alone select a unique measure.

Section 4 develops proposed comparisons, including whether variational laws can provide economical descriptions. No derivation of stationary action from shortest program length is claimed. Section 5 examines quantum interpretations and Section 7 addresses objections.

The position developed here (call it *mathematical monism*, or *Matheism*) extends the structural account of reality discussed in Ontic Structural Realism (OSR), including the accounts by Ladyman and Ross (2007) and French (2014, 2022). Standard OSR describes structure as reality's sole constituent but does not explain *why* structure exists. The rule-random dichotomy closes this gap: mathematical consistency is self-necessitating; consistent structures cannot fail to exist, just as 2+2 cannot fail to equal 4. The present paper develops the physical consequences. The Bell comparison retains local response and definite outcomes, then identifies the measurement dependence needed to reproduce the observed violations. A block account includes the apparatus within the explanation; it does not establish that dependence by itself.

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2. Foundation

Matheism identifies reality with mathematical structure. Its core argument asks what determinate features an allegedly nonmathematical constituent could have, and how it could interact without entering a common structure. Regularity and randomness can both be specified mathematically. An appeal to either therefore supplies no extra kind of substance.

The present physics account adopts that identity thesis. It distinguishes it from the stronger claim that every consistent mathematical structure is realized, and from the choice of a measure governing observations. Those commitments have different consequences and should not be counted as one proof.

Observers belong inside the same account as the systems they observe. A history explaining our evidence must recover their organization, their records and their physical interactions. The identity proposal does not need an additional consciousness-induced collapse mechanism; Section 4.7 states that point without making the philosophy of consciousness a second subject of this paper.

3. The Measure Problem and the Simplicity Dominance Principle

3.1 The Measure Problem

The Level IV multiverse resolves fine-tuning in principle but raises a new question: among all consistent structures, which should we expect to observe? Without a measure over structures, we cannot make predictions.

One might object that the question is meaningless: all structures exist equally, so there is no 'probability' of being in one rather than another. But this proves too much. If all structures are equally weighted, we should expect to find ourselves in a 'typical' structure. What counts as typical? Without a measure, we cannot say, but some measures yield absurd conclusions (e.g., under certain measures, almost all observers are Boltzmann brains with false memories).

The proposed answer is a description-based measure. In a fixed prefix-free binary language, shorter individual programs carry greater weight under fair-bit sampling. This gives a mathematical basis for the chosen prior. Its extension to physical structures and observer locations requires the representation and weighting rules stated below; the existence of structure alone does not select them.

I propose:

Algorithmic probability (Solomonoff, 1964; Schmidhuber, 2000): The measure of a mathematical structure S is governed by its algorithmic probability:

$$P(S) \propto 2^{-K(S)}$$

where K(S) is the Kolmogorov complexity of S, that is, the length of the shortest program that generates the structure's rules.

The chosen program measure must state its domain. A noncomputable infinite object lacks a finite program generating all its digits, but noncomputability is not identical to maximal algorithmic randomness. A computable stochastic law can assign total probability one to individually noncomputable sequences. Assigning zero probability to each singleton does not make their union probability zero. The domain of a particular program prior therefore does not prove that noncomputable structures have no ontological or observer weight.

Note that simple, finitely specifiable laws can generate non-computable phenomena as outputs (e.g., undecidable spectral gaps, chaotic systems requiring infinite precision). The laws have low K; some consequences may not. The measure weighs structures by the complexity of their generating rules, not their outputs. Theories with continuous dynamics but finite specification (such as string theory) remain consistent with SDP; the constraint is finite specifiability, which discrete approaches satisfy manifestly.

Critical distinction: K(S) measures *generative* complexity (the shortest program specifying the laws or rules), not *instance* complexity of specific states. A universe with simple laws (low generative complexity) can produce configurations of enormous detail, even infinite states. The Mandelbrot set has low K(S) (its definition is ~50 characters) but infinite fractal complexity.

One rule doing several jobs

The strongest economy would come from one construction explaining several features that would otherwise be specified independently: the laws, the special initial conditions and the particle content. A common geometric or dynamical rule is charged once; any extra choices needed for each result are still counted. This is a claim about shared explanatory information, beyond a preference for each ingredient to be simple on its own.

For example, suppose a specified code requires 100 bits for one account and another independent 100 bits for a second. A joint program of 130 bits that recovers both saves 70 bits against that 200-bit concatenation. Under the stipulated description weighting its individual weight is larger by 2^70. These numbers illustrate the comparison, not measured lengths of physical theories or a proof about their shortest programs. The physical task is to exhibit such shared derivations. That is why laws, boundaries and particle content belong in the comprehensive account.

3.2 Three Types of Probability

Three distinct concepts travel under 'probability' here; conflating them breeds confusion.

Quantum probability concerns credences within a branch. If Many-Worlds is correct, all measurement outcomes occur; the Born rule (that quantum probabilities follow the square of the wave amplitude) governs rational expectation about which branch contains one's continuation. This is epistemic: indexical uncertainty, not ontological indeterminacy.

Measure-theoretic weight concerns the distribution over mathematical structures. Simpler structures have higher measure under SDP. This is not about which structures 'get created' (all exist timelessly) but about their relative weight in structure-space, according to their program weights, not because shorter programs are more numerous.

Anthropic constraint concerns which structures can contain observers. Combined with measure, this yields predictions: observers should expect to find themselves in high-measure, observer-supporting structures.

These three probabilities are related but not identical. Quantum probability operates within a structure; measure-theoretic weight operates over structures; anthropic constraint filters the structures relevant to prediction.

3.3 Why SDP?

Four converging justifications support algorithmic probability:

Logical scope: A short hypothesis need not permit more possibilities than a long one. A compact rule can be highly restrictive, while a long specification can describe a broad family. Description length and the volume of allowed states therefore cannot be interchanged without a model relating them.

Measure-theoretic justification: In a prefix-free coding, each program of length k occupies a fraction 2^(-k) of program-space. A 10-bit program occupies fraction 2^(-10). A 1000-bit program occupies fraction 2^(-1000). The short program has 2^990 times higher measure, not because short programs are more numerous (they are not: there are more long programs than short ones) but because each short program claims a larger fraction of the space. If we imagine a universal Turing machine fed random bits, the probability of hitting any specific short program is exponentially higher than hitting any specific long one. This is Solomonoff's (1964) foundation for inductive inference; SDP extends it from epistemology to ontology. A "program" here is a finite description, not a process requiring execution. The structures exist as mathematics. The measure is over specifications, not computations. Schmidhuber (1997, 2000, 2002) developed this into algorithmic theories of everything; the present paper defends universal realization as an additional ontological premise. The coding measure specifies relative description weights; it does not itself establish that every described structure is physically realized.

Explanatory motivation: Compact physical laws motivate a search for short complete accounts. This observation does not define a uniform distribution over all structures or prove that observers overwhelmingly occupy simple ones. The relevant prior and observer weighting must be stated.

Invariance: Universal description languages agree on complexity up to additive constants. That property supports comparisons within a stated convention; it does not guarantee identical finite rankings or uniquely fix an ontological measure.

A potential circularity deserves address: algorithmic probability is defined relative to Turing machines, but Turing machines are themselves mathematical structures. Does this make algorithmic probability viciously circular, defining the measure over all structures in terms of one particular structure?

The equivalence of particular formal models of effective computation is mathematical. The Church and Turing thesis connects that formal class with the informal notion of effective calculation. Neither result determines a unique distribution over physically realized structures. Universal machines can share expressive power while assigning different finite description costs.

SDP is the measure proposal defended here. The burden is to state its advantages, assumptions and consequences, rather than require a critic to supply a

replacement before questioning its uniqueness.

Universal constructions retain choices of reference machine and representation. A probability measure within a physical state space and a prior over model descriptions answer different questions. Neither the absence of a uniform prior on one unbounded collection nor the existence of cosmological measure problems proves that a particular description prior is uniquely correct.

3.4 Occam's Razor as Metaphysical Principle

SDP provides a grounding for Occam's Razor. Traditionally, parsimony is treated as methodological preference: we should prefer simpler theories, but without clear justification for why simpler theories are more likely true. Hossenfelder (2018) pressed this point, asking why physics should prefer simplicity at all. SDP answers: lower-K structures have exponentially higher measure. The criterion is not beauty but brevity.

Under the specified weighting, a complete description that is shorter by Δ bits receives $2^{\wedge}\Delta$ times the initial weight. That is the exact simplicity result. Explaining the success of simple physical laws also requires justifying the weighting and showing how the candidate descriptions predict our observations.

The compact form of successful physical laws motivates this proposal. A short law can generate complicated phenomena. A complete comparison must nevertheless include its starting conditions and observational predictions, so that apparent economy in the equation does not conceal information elsewhere.

3.5 Limitations and Open Questions

Algorithmic probability faces technical challenges that deserve acknowledgment:

Non-computability: Kolmogorov complexity K(S) is uncomputable; there is no algorithm that takes a structure and outputs its complexity. This limits our ability to calculate exact measures. However, we can often establish bounds (a structure's complexity is at most the length of any program that generates it) and make comparative judgments.

Machine-dependence: Universal languages agree on Kolmogorov complexity up to an additive constant. That constant depends on the languages and can change a finite ranking. A 100-bit versus 200-bit difference gives a factor $2^{\wedge}100$ within the specified weighting; it is not guaranteed to survive an arbitrary change of language.

Measure underdetermination: Under Lebesgue measure on [0,1], algorithmically random reals have measure 1; under SDP, computable reals dominate. Both measures are mathematically legitimate. This may represent a fundamental limit: the 'correct' measure over mathematical structures might be metaphysically underdetermined. If so, Algorithmic probability is not the unique

solution but one principled choice among several. The underdetermination parallels gauge freedom in physics: multiple mathematically legitimate choices, but some more natural than others. Algorithmic probability is the 'natural gauge', the choice singled out by independent theoretical virtues. I flag this as an open problem rather than a defeater.

Argumentative status: Given the a priori foundation, the Level IV multiverse follows deductively: all consistent structures must exist if existence is mathematical consistency. *Which measure* governs the ensemble is not derivable a priori but is established by inference to best explanation. Algorithmic probability is the best available hypothesis for the measure problem, with theoretical virtues no competitor matches.

Despite these challenges, SDP provides a principled basis where alternatives offer none.

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4. Consequences and Predictions

4.1 Explaining fine-tuning

Fine-tuning arguments ask how changes to specified constants affect structures such as atoms, stars and galaxies. The answer depends on which constants vary together and which physical requirements are imposed. A single universal percentage beyond which chemistry fails is not established here. The explanatory comparison must state the candidate variations and recover the relevant physics.

Standard responses invoke either design (theism) or selection (weak anthropic principle within a multiverse). Algorithmic probability offers a third option: the constants are not 'tuned' at all. Among all observer-supporting structures, those with simple laws dominate. Simple laws produce orderly universes where constants take values permitting complexity. Fine-tuning is what simplicity looks like from the inside.

Computable constants such as pi have algorithms generating arbitrary finite precision. The description cost includes the chosen language and, for a finite requested output, its precision or termination rule. No language independent twelve bit cost is established here. The search for equally economical physical parameter relations remains worthwhile, without assuming that every parameter must have one.

More precisely: let S_obs be the set of observer-supporting structures. Algorithmic probability predicts we should find ourselves in a structure with low K(S) relative to S_obs. This is consistent with our universe having elegantly simple fundamental laws despite the apparent fine-tuning of constants.

4.2 The Variational Form of Physical Laws

A quantum path integral weights contributions with action dependent complex phases. In a suitable semiclassical regime, rapidly varying phases can suppress contributions away from stationary action, yielding a classical approximation. Several stationary paths can contribute, and corrections need not vanish. Classical action is stationary, not necessarily minimal.

This mechanism uses more than the existence of a map. Its action, amplitudes, integration prescription and limiting regime do the work. Ordinary averaging over arbitrary discrete updates has none of those ingredients automatically. Comparing an action formulation with an exhaustive list of trajectories also gives it an unfair advantage over other compact generators.

Noether's theorem relates appropriate continuous symmetries of a variational problem to conservation laws under its hypotheses. It does not prove that stable structures require an action formulation. Starting with observers, demanding stability and citing Noether does not derive an action that the theorem already presupposes.

A proposed universal connection between observers and variational dynamics remains a separate conjecture. Its proof would need a definition of observer, a candidate class and a result excluding alternatives. Familiar atomic chemistry is evidence about this universe, not a proof that every possible observer uses identical chemistry.

4.3 The Many-to-One Root

A proposed discrete account needs an explicit map from its microscopic states to effective observables. The induced evolution may have memory, noise or dependence on discarded variables. A closed effective law requires justification.

A successful candidate should recover a specified metric or causal structure, the appropriate symmetries, quantitative propagation and interaction laws, and observation probabilities. Branch reconvergence by itself does not provide complex amplitudes, destructive interference or the Born rule. Claims about general covariance likewise do not alone establish Einstein's field equations or their measured solutions.

4.4 Testable Predictions

The simplicity proposal favors complete accounts that explain the same observations with fewer independent inputs. That supplies a concrete target for unification: derive parameter relations, particle content or effective geometry that a rival must specify separately. A large output is compatible with a short generator.

Under a fixed prior proportional to 2^(−L), an established k-bit saving gives an individual description 2^k times the prior weight. Posterior comparison

also includes observational likelihoods. A less accurate model can lose despite its shorter code. The proposal supplies no universal maximum size for a final theory.

A successful discrete account must derive the physical quantities being compared, including the effective geometry, propagation and observation probabilities. A short hypergraph rule is a candidate generator, not already a derivation of the Standard Model. A controlled map to a continuum approximation would establish the relevant large-scale predictions; neither discreteness nor shortness guarantees that map.

The evidence that could favor the proposal is repeated, matched success: previously independent observations following from one economical construction. A comparison that omits boundary information, uses unmatched software packages or changes its observer measure after seeing the result would not establish that success.

4.5 String Theory and the Complexity Test

Under mathematical monism, all string vacua exist without selection. But low complexity of a theory itself is necessary and not sufficient: pi and Wolfram's hypergraph rules are also low-K, and neither has been shown to reproduce our physics. The theory that explains the most with the least specification wins.

4.5.1 Parameters as Addresses: A Caveat on the Conservative Count

An address description is available when an explicit generator and selector reproduce the required model. It gives an upper bound on K, not an equality or a guaranteed saving. The earlier 390 bit example was 26 values at 15 bits each; it was not a verified total for nineteen parameters or complete physics.

The completeness theorem for classical first order logic connects syntactic consistency with existence of a mathematical model. It does not establish physical realization. Matheism's stronger realization thesis requires its own ontological argument.

The information-theoretic point: log_2 of the number of distinguishable observer-compatible regions in the gauge-theory space is an upper bound on the address cost. If the space contains, say, 10^20 such regions, the address is roughly 67 bits. The parameter values that distinguish our region from neighboring ones are consequences of the region, not independent free choices.

4.6 Emergent Geometry

A short generator could yield large-scale geometry. This paper does not derive that geometry from description length.

4.7 Consciousness within the mathematical ontology

A proposal that entanglement or wavefunction collapse produces consciousness would still specify mathematical states and relations. It therefore adds no non-mathematical ingredient. The further claim that this particular mechanism occurs and constitutes experience needs its own derivation or evidence. The identity argument does not need that mechanism and does not become stronger merely by invoking it.

5. Quantum Mechanics and Many-Worlds

5.1 The Wavefunction as Mathematical Structure

The quantum wavefunction Ψ is itself a mathematical object: a vector in Hilbert space, evolving unitarily under the Schrödinger equation. On the present view, this is not a representation of something else; it is the thing itself. The wavefunction does not describe reality; it *is* reality at the quantum level.

The identity proposal places the quantum state within mathematical reality. A measurement is then an interaction between physical systems. A Hermitian observable has real eigenvalues, representing its possible measured values; the operator itself generally maps vectors to vectors. Interaction with the environment can suppress interference between stable records, the process called decoherence (Zurek, 2003). In an Everettian account these records belong to branches of the joint state. Recovering their probabilities and explaining the observer's recorded outcome remain parts of the interpretation. Decoherence alone is not a physical collapse selecting one global outcome.

5.2 Many-Worlds as Natural Interpretation

The Many-Worlds Interpretation (MWI) takes the Schrödinger equation literally: all branches exist, and apparent 'collapse' is subjective, with each observer-branch seeing one outcome. MWI coheres naturally with the present position:

Parsimony: Everett avoids a physical collapse mechanism. A fair comparison must still specify its state, Hamiltonian, boundaries and observational weights alongside the corresponding commitments of competing accounts.

The Schrödinger equation alone is not a complete predictive model. Objective collapse models add specified stochastic dynamics; Bohmian models include a configuration and guidance rule. Comparing complete descriptions under one coding convention can assign a prior advantage to a shorter account. No minimum K ranking follows merely from this inventory.

Predictive completeness: A finite random outcome has finite description length. A stochastic law supplies a likelihood without listing each outcome as part of the law. An Everettian account must also explain the weights of the records observers encounter. This separates the description of a law from the exact specification of a history and removes the claimed infinite complexity gap.

Relation to universal realization: If all admissible mathematical structures are realized, one can consider structures with different dynamics. This premise does not establish which quantum account describes our observations or what observer weighting to use within it.

The prior comparison remains open until complete candidates and likelihoods are specified. The universal realization claim and a ranking of observer expectations are distinct commitments.

A stochastic model can have a short law even when a typical individual outcome sequence is incompressible. Its outcome probabilities enter the likelihood. A deterministic model must likewise account for initial conditions and observed records, and an Everettian model must connect branch weights to observations. Charging random outcomes only to collapse models while charging no observational selection to rivals does not provide a fair comparison.

5.3 Bell's Theorem as Corroboration

Bell inequality violations constrain models satisfying local factorization, measurement independence and the other assumptions of the experiment. They do not independently establish mathematical structural realism or exclude every deterministic account.

Different quantum accounts revise or interpret those assumptions differently. Replacing this range of choices with a dilemma between nonlocality and abandoning reality obscures the actual model comparison.

A structural account includes spatially separated systems and their correlations in one description. That does not remove the need to recover Bell statistics or distinguish controllable signalling from locality in the underlying model. Spatial and temporal relations are part of the structure, but not every arbitrary relation is a physical distance or transition rate.

5.4 The Born Rule

The Born rule assigns probabilities from squared amplitudes. For a normalized state $|\Psi\rangle$ and a projector P onto an outcome subspace, the assigned probability is $\langle\Psi|P|\Psi\rangle$. Hilbert-space geometry gives this expression a precise mathematical role. Showing why observers should use it as a probability rule still needs the assumptions of the proposed derivation. Deutsch (1999) and Wallace (2012) develop a decision-theoretic route in Everettian quantum mechanics; this paper does not replace that work with a description-length argument.

Algorithmic probability does not independently derive the Born rule. A code constructed from Born weights already uses the probabilities it would supposedly explain. No general result here equates a branch's Kolmogorov complexity with the negative logarithm of its squared amplitude.

An Everettian comparison must state how it obtains observational weights and

stable records. The research preference for a unitary account is retained, while the claim that SDP has already selected it is not.

5.5 Reversible quantum evolution and records

A record can omit details even when the complete dynamics retains them. Under unitary quantum evolution, information inaccessible within one subsystem can remain in correlations with its surroundings. Local loss of accessible information therefore does not alone require wavefunction collapse.

This also explains why a many-to-one description, which gives the same output for different inputs, need not imply irreversible microscopic laws. The direction in which reliable records develop depends on the history of the system and environment. Unitary dynamics alone supplies neither a preferred temporal orientation nor an observer-weighting rule.

5.6 Superdeterminism Rehabilitated

A proposed algorithmic prior can compare deterministic and stochastic predictive models when the same explanatory task and coding convention are used. Neither class wins by definition.

Specificity does not imply a long generating description. That answers one inference sometimes associated with fine tuning, but not objections about causal faithfulness, sensitivity or preparation measures.

For predictive comparison, describe the complete model M and evaluate P(D | M) for the observations D. Joint encoding can exploit dependencies; K(L,I,O) need not equal K(L) + K(I) + K(O). Random outcome records are data, not automatically extra law complexity.

This distinction does not make a mathematical proof depend on experiment. A proof of a model's predictions, or of a weight ratio under a specified prior, is deductive. The a priori Matheism argument concerns the justification of structural ontology; choosing its particular physical realization is an additional task unless uniquely derived. Empirical evidence tests the application of a model, not the validity of a correct mathematical implication.

A short common generator could explain specific measurement-dependent correlations. It must also recover the correct statistics. For example, a counter can cycle over balanced binary settings a,b and a shared bit r, with outputs A=r and B=r XOR (a AND b), where the shared source supplies the information used in the latter response. This gives balanced one-wing marginals and a CHSH value of 4, above the quantum bound. Shortness and absence of observable signalling therefore do not alone recover quantum physics.

Ciepielewski, Okon and Sudarsky discuss homogeneous preparations and a modified relaxation proposal, with unresolved dynamical issues. This is not a general

proved attractor theorem. Palmer proposes a different geometry for admissible states; a metric alone does not supply a probability measure.

An Everettian account, a measurement dependent model and a stochastic collapse model must each include their observation rule. A shorter complete account with equal likelihood receives the corresponding prior advantage; a less accurate account can lose that advantage through the likelihood. The comparison has not been numerically completed here.

The research task is to compare complete generators and predictive models under common standards. An automatic collapse penalty for unbounded outcome complexity does not perform that comparison.

The Deterministic Constructible Universe Principle is a proposed restriction to physically realized histories with deterministic constructive accounts. It is not implied merely by the existence of mathematical structure: stochastic transition laws are also mathematical. A computable distribution can have typical infinite realizations without finite deterministic generators. Defending the restriction requires an additional ontological argument, not equating randomness with inconsistency. Deterministic continuum theories also need constraints on their initial data to guarantee computable histories.

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6. Particle Content: What a Compact Unification Must Explain

The observed gauge structure, generations and parameters are separate targets for the proposed common account. Explaining one does not automatically explain the others.

Gauge groups and shared structure

The Standard Model's gauge group is SU(3)_color × SU(2)_left × U(1)_hypercharge. To specify it requires encoding three pieces of information: which compact Lie groups, in what dimensions, with which representations carrying which fields.

Exact Kolmogorov complexity is not generally computable. Constructive encodings provide upper bounds. A shorter known encoding on one side does not prove an ordering of the true minima unless the other side is bounded appropriately.

Consider the candidates that have been proposed as either replacing or unifying the Standard Model:

SU(5). A unified group can reuse representations while introducing a symmetry breaking sector. Physical predictions, including proton decay, depend on the

specified model. Neither the group name nor the existence of experimental constraints supplies its total description cost.

SO(10). Its representation structure can organize a fermion generation compactly. The breaking sector, interactions and predicted observations still need specification. No net K ranking relative to the Standard Model is calculated here.

Pati Salam models. Quark and lepton unification and a right handed sector can introduce both explanatory relations and additional ingredients. Their balance must be evaluated in an actual model.

Heterotic constructions. A larger group can have a short mathematical definition. Obtaining a particular low energy spectrum and its dynamics is the relevant physical task. Group dimension alone supplies no description length penalty.

Trivial gauge content. It fails to reproduce observed interactions. The stronger claim that no imaginable observer could exist under any theory with that content requires an independent definition and analysis of observers.

Smaller gauge sectors. Failing to reproduce our observed particle physics is not the same as proving that every alternative form of organized matter is impossible. Observer compatibility must not simply be defined as possessing the Standard Model one intends to derive.

An observer conditioned comparison can impose independently justified physical requirements. It cannot assume our full gauge structure and then report that structure as the selection result.

No monotonic relation between group size and K has been established. A larger structure can have a shorter generating rule. Demonstrating a local minimum requires explicit competitors and costs.

Why three generations?

The Standard Model contains three generations of fermions: (u, d, e, _e), (c, s, , _), (t, b, , _). Each generation duplicates the structure of the first but with higher masses. Why three?

Experiment establishes the observed generation structure and constrains particular extensions. Such constraints are model dependent. Counting neutrinos with specified masses and couplings is different from ruling out every additional fermion sector.

The connection to CP violation gives a restricted structural fact, not a selection of all observer compatible theories.

For N nondegenerate quark generations in the usual charged current framework, the number of physical CKM phases is (N−1)(N−2)/2. Three generations are the minimum for that mechanism to contain a CP violating phase. A two

dimensional unitary matrix can have complex entries; the point is that its phases can be removed by the allowed field redefinitions.

Allowing a CKM phase is not sufficient for successful baryogenesis. The Particle Data Group's CP review states that the observed Kobayashi Maskawa mechanism falls far short of the cosmic asymmetry. Other sources and mechanisms remain under investigation. Thus an inference from CKM phases to a minimally sufficient observer producing cosmology fails.

The restricted comparison is:

One and two generations have no physical CKM phase in this framework. Three allow one; additional generations allow more. This counts phases under stated field assumptions. It does not prove which generation counts permit observers across all physical models.

The remaining selection task is explicit. Such a result would require competing complete cosmological models, including mechanisms that produce the needed matter abundance.

The observed count motivates the comparison. It is not a retrospective prediction until the premises and calculation actually select it.

A proposed alternative baryogenesis sector adds ingredients but may explain other observations. Its total cost must be compared with a successful competitor. The observed CKM mechanism cannot be labeled the cheapest successful route when it does not produce the required asymmetry.

Masses, couplings and numerical relations

Parameter totals must specify whether neutrinos are massive, whether their masses are Dirac or Majorana, and whether gravity and cosmology are included. Listing ingredients from several extensions while calling them the nineteen parameters of the minimal model is inconsistent.

A short formula for measured parameters would be valuable if it predicted them to the required accuracy with fewer independent inputs. The formula and its domain must be stated.

Approximating a measured angle by a small rational expression after inspecting the value does not establish a physical law. Choosing among many possible expressions itself has a cost. Tests should use held out predictions, precision requirements and an explicit search procedure.

The mass hierarchy and coupling values remain explanatory targets. Running quantities also depend on scale and renormalization convention, which must accompany any numerical relation.

The prior does not establish that the full observed parameter assignment lies near a universal minimum. It defines one proposed preference among complete descriptions after the candidate class and observational task have been specified.

A deeper theory with fewer independent inputs could improve compression, including a theory developed within string theory. A claim of improvement must count its full rule, boundaries and prediction error. SDP supplies no blanket theorem excluding that possibility.

Parameters as addresses

The earlier 390 bit budget was 26 times 15. It was not a measured cost for nineteen parameters or all physical laws. Specifying precision needs units, ranges and a code. Treat parameter budgets as stated examples until those choices are supplied.

Given a computable generator G and an address a that actually yields a model T, K(T) K(G)+K(a)+O(1). This is an upper bound, not an equality. An alternative direct description may be shorter. Both routes deserve the same treatment across competing theory families.

For 10^20 agreed alternatives, a fixed length address uses 67 bits given their enumeration. The generator and observation rule still cost something. The number is illustrative; no catalogue of that many observer compatible physical models has been constructed here.

Calling a space primordial does not provide its generator or select our location. SDP weights completed descriptions; it does not construct every consistent gauge theory or identify observers automatically. A short selection rule within a landscape must be credited on the same basis.

Repeated agreement of measured constants is not proof of exact infinite precision or of an address mechanism. A discrete underlying model can also have exact symmetries and stable effective constants. Any prediction of observable fluctuations requires a specific microscopic dynamics and scale; it does not follow from discreteness alone.

The address proposal earns a shorter description only when its generator and selector reproduce the required physics more economically than a direct model under the same conventions.

No total bit count or exponential superiority over a theory family is retained as a result.

7. Objections and Responses

7.1 'This Is Untestable Metaphysics'

Objection: The Level IV multiverse is empirically inaccessible. Other structures cannot be observed. This is metaphysics, not science.

Response: Empirical assessment requires a specified predictive model. A simplicity prior that gives a complex model nonzero weight is not logically refuted

by one complex outcome, and universal algorithmic results do not require every law to have a variational formulation. Models motivated by the prior can be compared through predictions; the ontological extension remains a separate claim.

Moreover, many accepted physical theories posit unobservable entities. Quarks cannot be isolated; the wavefunction cannot be directly measured; some aspects of very early cosmic history are inferred indirectly. Science routinely posits unobservables that explain observables. The Level IV multiverse is in this tradition, posited because it explains what we see (fine-tuning, simple laws, mathematical effectiveness).

Quantum computing tests predictions of quantum dynamics. Its success alone does not discriminate Everett from other accounts that predict the same computation. An experiment distinguishes models when their observational predictions differ.

7.2 'The Measure Is Ad Hoc'

Objection: Algorithmic probability is chosen precisely to yield desired conclusions. Why not some other measure?

Response: Universal algorithmic constructions have dominance properties for specified effective classes. These motivate a prior without defining a unique reference language or a unique physical measure. Those additional choices must be stated.

Parsimony is a useful guide when assessed alongside observational fit. Its usefulness motivates testing a simplicity based method; it does not prove that the proposed ontological measure governs existence.

7.3 'The Measure Problem Is Unsolvable'

Objection: Any proposed measure faces the question 'why this measure?' We have merely pushed the problem back one level.

Response: This is partly correct. Algorithmic probability does not eliminate all 'why' questions; it relocates them. We trade 'why these physical laws?' for 'why algorithmic probability?' The latter is arguably more tractable: algorithmic probability has unique mathematical properties (dominance over computable measures, invariance under machine choice up to constants) that physical laws lack.

Some explanatory terminus is necessary. Every account reaches a point where further 'why' questions become incoherent. The question is whether we terminate at a principled stopping point or an arbitrary one. Algorithmic probability terminates at the structure of computation and information, fundamental concepts that may not admit further reduction.

7.4 Anthropic Challenges

Objection: Algorithmic probability faces multiple anthropic challenges: Boltzmann brains, the simulation argument, the Fermi paradox, and our temporal location. How does the position address these?

Response: A prior over candidate histories and a rule for locating observers within each history are distinct. The Conditional Parsimony Postulate proposes a description based location rule. That rule must be defined on a specified domain and checked against the physical observation probabilities; it cannot silently replace them.

Two Levels of Complexity. Addressing anthropic challenges requires distinguishing two measures:

Level 1: candidate histories. One proposed prior assigns weight proportional to 2^(−K_U(S)) to effectively described candidates S. A complete candidate includes its law, boundary conditions and any further information needed to identify the history. The complexity of the laws alone does not distinguish all the histories those laws permit.

Level 2: observer locations. For a specified finite or countable class of effectively described locations j within S, CPP proposes:

q(j | S) 2^(−K_U(j | S)).

This is a proposed normalized location prior on that class, using a fixed conditional coding convention. It is not identical to the Born rule, a thermal distribution or a probability over every exact continuum microstate. Its empirical adequacy and relationship to physical measures require an argument.

7.4.1 Boltzmann Brains Some cosmologies permit fluctuation observers that could dominate a proposed observation measure. The concern is model dependent; eternal duration alone does not create it.

For example, a constant positive formation rate integrated over an unlimited duration gives an unbounded expected event count. If ordinary observer production is finite and the adopted measure weights those events comparably, fluctuation observers can dominate. A decreasing rate with finite integral, a dynamical obstruction or a different observation measure changes that conclusion. Each premise must be established for the proposed cosmology.

Infinite duration alone does not establish the formation or dominance of Boltzmann observers. Determinism does not exclude ergodicity: a finite cyclic counter visits every state. The relevant cosmology must supply accessible histories with sufficient organization for experience and a measure for their observations. Boddy, Carroll and Pollack's stationary vacuum argument and Tumulka's Bohmian field analysis concern particular dynamical assumptions. They are not instances of a theorem excluding fluctuation observers from every deterministic universe.

A formation argument must identify the relevant histories and their weights. A nonzero instantaneous projection onto a brain shaped configuration does not alone demonstrate a functioning observer. This distinction is already present in the stationary vacuum and Bohmian literature.

Description length does not provide an independent suppression theorem here. A simple cosmological generator may produce a fluctuation observer as well as a developed observer. Both candidates must be allowed generative descriptions, with consistent charges for location, time and observational likelihood.

Implication: the question remains open for a specified cosmology until its dynamics, observer criterion and measure are supplied.

7.4.2 Simulations A simulation argument compares the weights of observers implemented in simulations with other observers. Large possible populations motivate the comparison but do not calculate its probabilities.

If a complete history generates a computer and its program, do not charge that machinery a second time as independent information. Conversely, laws permitting computers do not guarantee that any particular computer or simulation develops. The relevant description includes whatever the law and boundary fail to determine, together with the observer location rule.

Implication: neither decisive suppression nor dominance of simulated observers has been established.

7.4.3 Fermi Paradox and Alien Colonizers Models of interstellar expansion require separate physical and observational inputs.

Physical law: a candidate must permit the transport, energy use and computation needed for the proposed expansion. Those capabilities may already follow from the same laws supporting local observers. There is no established extra law complexity merely because a society travels farther.

History: technology and expansion may follow from a compact generator or require additional contingent information. Their description cost cannot be inferred just by listing the apparent intricacy of a civilization. Compare complete generators and, where appropriate, likelihoods of the outcomes.

Observation: nondetection depends on prevalence, timing, detectability and the search performed. A subset of branches need not have a small total Born weight, and the phrase Great Filter supplies no numerical rate by itself.

Expansion could also change observer counts and locations. A calculation must state how those changes affect the evidence being conditioned on.

Implication: a useful model joins expansion dynamics, detection likelihoods and observer weighting. The present simplicity proposal is a framework for that comparison, not a solution to the Fermi question.

7.4.4 Temporal Location: Why Early Rather Than Late? Elapsed time, entropy and the complexity of an observer's address are different quantities. A very large integer can have a short description, and a late state can be generated from compact laws and a boundary. Branch number does not by itself give observer weight. Specify the code and sampling rule before comparing early and late observers.

A model with many later observers makes late location typical under uniform sampling within that history. Other location measures can give another result, and late simulations may contain records of an early era.

Implication: our records can inform a defined model of histories and locations. SDP alone does not predict that we should appear early or close to a transition.

7.4.5 Information, Development and the Arrow of Time The proposed arrow concerns the production of information. In a history with a first event and no last, forward generation can begin with supplied information. A reverse dependency chain in which every step waits for a later step cannot start without an independently supplied input. This excludes that reverse construction. Connecting it to experienced time requires showing why experience needs such production, rather than the relevant relations simply existing together in a mathematical block.

If a history starts without the organization its later observers require, that organization must develop. The laws themselves can remain simple. In biological histories like ours, available energy supports that development while total entropy increases. Neither organization nor entropy must increase at every step.

Coordinated convergences in reversed splashes may depend on details a local learner cannot sense. If the same available evidence hides situations requiring incompatible responses, no decision based on that evidence can guarantee success in both. Later observations may explain the event, too late for the earlier choice. Repeated dependence on such inaccessible information could constrain the evolution of reliable learners. A directional preference must survive a comparison of observers and environments together: full reversal also transforms the records.

Start, hypothetically, with neither temporal direction favored: a coin flip. The proposed arguments then favor forward development through rooted information production, the development of initially absent observer organization and reliable evidence available before action. Even if reversal is possible, it could receive less probability weight under the stated measure. Establishing that preference requires showing that additional specification or restrictions on viable observer histories actually reduce their total weight, including comparison with a fully reversed observer and environment. The arguments share assumptions and cannot be counted as independent probability gains.

This developmental question connects to work on robust memory: Mlodinow

and Brun (2014) argue for alignment of the memory and thermodynamic arrows under explicit robustness conditions, including for reversible recorders. It supplies a possible route from physical histories to observers' temporal orientation without requiring a microscopic erasure law.

A demonstrated extra description cost would supply another preference. Under the same weighting by $2^{\wedge}(-\mathrm{L})$, where L is the complete description length in bits, five extra bits reduce individual weight by a factor of 32. A reversal instruction added to one program does not prove that every reverse description needs it. A compact rule can specify coordinated events.

A short description does not by itself imply low thermodynamic entropy or usable energy differences. The explanation must include the starting conditions, dynamics and the weight assigned to histories with the relevant observers. That is where simplicity can support the proposed physical selection.

7.4.6 Epistemic Status Summary The comparisons share an ontology but retain distinct missing inputs.

Simulations and temporal location: implementation, population histories and observer weighting are required. Computers may develop under simple laws without adding independently specified machinery, but that development and the simulations they run cannot be assumed in every candidate history.

Fluctuation observers: calculate the accessible histories, determine whether they satisfy the experience criterion and compare their observational weight. The simplicity prior needs to be combined with those ingredients on equal terms.

An observer criterion based on temporal organization cannot be reduced to counting formal updates. One update can abbreviate an elaborate process, while another can be a trivial bit flip. If a fluctuation implements sufficient organization it should be treated consistently with any other implementation. Whether it does is a further question, not a fact established by calling it one conscious tick.

The physical comparisons here assume the stated structural ontology. They do not independently resolve every objection to that ontology.

8. Conclusion

The aim is one economical account of physical laws, geometry, quantum observations and the histories in which observers develop. Under the stated simplicity rule, a genuine saving in a complete description earns an exact advantage in initial weight. Establishing that saving requires the same predictive task, precision and coding conventions for each candidate.

Quantum evolution without collapse offers one economy: the same rule applies to systems and observers. Superdeterminist models pursue another: a common physical cause for the particle source and the measurement settings. Each must

account for the observed probabilities under its own assumptions. A beginning that lacks observers must develop the organization they require. The proposed temporal explanation combines a source for information production with conditions that support reliable learning and adaptation. Its physical application requires an account of which histories satisfy those conditions and why observers encounter them.

These are connected parts of the physical explanation. The common mathematical foundation motivates looking for shared rules that do several jobs. A successful unification must display those rules and recover the observations, making clear which results follow by proof and where comparison with our universe enters.

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